# Process Characterization Master Plan

**A-Mab Drug Substance — PCMP-001**

Process Development / Manufacturing Science & Technology (MSAT)

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## Approvals

Table 1: Approval record (synthetic; signatures not applied).

| Role | Function | Name (synthetic) | Signature | Date |
| --- | --- | --- | --- | --- |
| Author | Process Development / MSAT | — | — | — |
| Reviewer | Analytical Development | — | — | — |
| Reviewer | Manufacturing Science | — | — | — |
| Reviewer | Statistics | — | — | — |
| Approver | Quality Assurance | — | — | — |

## Abbreviations

ADCC, antibody dependent cell mediated cytotoxicity; AMV, analytical method validation; CDC, complement dependent cytotoxicity; Cpk, process capability index; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; KPP, key process parameter; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; OLS, ordinary least squares; PPQ, process performance qualification; PRESS, predicted residual error sum of squares; SEC, size exclusion chromatography; SOP, standard operating procedure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

# 1 Purpose and scope

This master plan governs the Stage 1 characterization of the A-Mab drug substance process. It covers the 8 unit operations of the drug substance train, from the production bioreactor through to ultrafiltration and diafiltration (Steps 3 to 10). All studies are executed on qualified scale-down models at the sending site (Novacyte Biologics — Cambridge, MA (Development)), and their purpose is to establish the operating ranges, the proven acceptable ranges and the control strategy under which the process will run at the receiving site (Novacyte Biologics — Grafton, WI (Commercial DS)) after transfer.

The plan exists so that the individual characterization plans do not each restate the same framework. It states once, for the whole campaign, which quality attributes the process must control, how the risk assessment scoped the studies, how scale-down models are qualified, how the data are analysed, and what the acceptance criteria are. Each characterization plan then carries only what is specific to its own unit operation. Where a step-specific matter is treated in both documents, the characterization plan governs.

The campaign covers 37 process parameters and 10 drug substance quality attributes. The commercial process runs at a 15,000 L production bioreactor working volume and delivers drug substance at a target concentration of 75 g/L, and every range studied in the campaign contains the set-point of that process.

This plan sits inside the transfer package described in PTP-001, which is its parent (Table 1.1). RA-001 provides the risk basis, and its study assignment is the input to every characterization plan. The characterization plans PCP-003 to PCP-010 are the children of this plan and are registered in §8. The reports they produce are consolidated in PCMR-001, which is the document that states whether the process is ready for Stage 2.

Table 1.1: Parent documents of this master plan and the report the campaign rolls up into.

| Document ID | Title | Relationship |
|---|---|---|
| PTP-001 | Process Transfer Plan (A-Mab Drug Substance) | Parent — transfer scope |
| RA-001 | Pre-Characterization Process Risk Assessment (A-Mab Drug Substance) | Parent — risk basis for study scope |
| PCMR-001 | Process Characterization Master Report (A-Mab Drug Substance) | Rolls up into |

Several activities are outside the scope of this plan. Drug product manufacture and its characterization are covered elsewhere. Validation of the analytical methods is covered by the method validation documents listed in Table 5.1, and the campaign uses those methods as validated. The execution and reporting of the Stage 2 process performance qualification belongs to a separate protocol under PTP-001. The design of each individual study is also outside this plan, because it is set in the characterization plan for the step.

# 2 Stage 1 characterization strategy

The campaign follows the enhanced development approach of ICH Q8 inside the lifecycle model of the process validation guidance (International Council for Harmonisation 2009; U.S. Food and Drug Administration 2011). Prior knowledge and a formal risk assessment identify what has to be studied, characterization then bounds the process over ranges wider than routine operation, and the outcome is a control strategy that holds each parameter where the data support it and monitors the attributes those parameters govern. The A-Mab case study is the methodological model for the whole package (CMC Biotech Working Group 2009).

Stage 1 of the lifecycle is process design, and these studies are its record (U.S. Food and Drug Administration 2011). Characterization is a set of documented studies in which process parameters are varied on purpose, so that their effect on quality attributes and on process performance can be measured (International Council for Harmonisation 2012). The studies are not a demonstration that the process works. They are the evidence from which the ranges, the classification of each parameter and the in-process controls are derived.

Criticality is treated as a continuum and not as a binary label (International Council for Harmonisation 2023b). Each attribute carries a criticality level from the impact and uncertainty assessment described in §3, and the attributes at the high and very high levels are called critical. The same principle applies to parameters. A parameter is classified from the size of its effect on a critical quality attribute and from the risk that it moves outside the region in which that attribute is controlled.

The ranges studied define the knowledge space, which is deliberately wider than the region the process will run in. The design space is the part of the knowledge space in which every governed attribute stays inside its acceptance criterion. The normal operating range sits inside the design space and is where the process is held in routine manufacture. Ranges are studied beyond the normal operating range for two reasons: an excursion outside the routine range can then be assessed against data instead of against judgement, and a later process improvement has a basis that does not need a new characterization campaign.

The control strategy that this campaign feeds has four elements. The first is the set of parameter ranges and the classification behind each of them. The second is the in-process monitoring and the in-process limits at the points where a deviation can still be caught. The third is release testing of the drug substance against the criteria in Table 3.1, which is the final confirmation that each attribute is where the process design says it should be. The fourth is the set of lifecycle controls, such as resin and membrane use limits, buffer and feed controls, and the change control that governs movement inside and outside the design space. Each characterization plan states which of these elements its step is expected to contribute to, and the report for the step states what it in fact contributed.

Prior platform knowledge carries a large part of the argument. A-Mab is produced on a humanized IgG1 platform that has supported three licensed antibodies (X-Mab, Y-Mab and Z-Mab) and several clinical programmes. The mechanisms by which the platform steps act on quality attributes are established, so this campaign confirms and bounds platform behaviour instead of discovering it. Where prior knowledge is transferred to A-Mab, the plan for the step states the mechanism that makes the transfer valid, and does not rest on similarity alone.

The plan and the report for a step deliver, between them, a characterized region for the parameters studied, proven acceptable ranges for the parameters that govern an attribute, a classification for every parameter of the step, and a statement of what the step contributes to the control strategy. Characterization on scale-down models does not qualify the commercial process. That is the purpose of Stage 2, in which 5 process performance qualification batches are run at commercial scale under a separate protocol.

# 3 Critical quality attribute framework

The process must control 10 drug substance quality attributes. They are given in Table 3.1 with their acceptance criteria, their criticality and the step the register names as governing each of them.

Table 3.1: Drug substance quality attributes in scope of the campaign, with acceptance criteria, criticality (Tool #1 score) and the governing process step.

| CQA | Category | Acceptance | Criticality | Tool #1 | Governing step |
|---|---|---|---|---|---|
| Afucosylation | glycosylation | 2–13 % of glycans | H | 60 | Step 3 — Production Bioreactor |
| Galactosylation (total %Gal) | glycosylation | 10–40 % (G1+G2 equiv.) | H | 48 | Step 3 — Production Bioreactor |
| High Mannose | glycosylation | 3–10 % of glycans | VH | 80 | Step 3 — Production Bioreactor |
| Aggregates (HMW) | size variant | 0–5 % HMW (SEC) | H | 60 | Step 3 — Production Bioreactor |
| Acidic charge variants (deamidation) | charge variant | 18–40 % (CEX/icIEF) | VL | 4 | Step 3 — Production Bioreactor |
| Host Cell Protein (HCP) | process impurity | 0–100 ng/mg | M-H | 36 | Step 3 — Production Bioreactor |
| Residual DNA | process impurity | 0–0.001 ng/dose | VL | 6 | Step 3 — Production Bioreactor |
| Leached Protein A | process impurity | 0–5 ppm | L-M | 16 | Step 5 — Protein A Chromatography |
| Viral clearance — XMuLV (enveloped) | viral safety | 16.7–30 log10 (cumulative) | VH | 20 | Step 6 — Low-pH Viral Inactivation |
| Viral clearance — MVM (parvovirus) | viral safety | 8.6–20 log10 (cumulative) | VH | 20 | Step 8 — Anion Exchange Chromatography |

The register carries 6 attributes of high or very high criticality, and those are the ones treated as critical (CMC Biotech Working Group 2009). Criticality follows the impact and uncertainty assessment of the case study (Tool #1), in which the score is the product of the impact of the attribute on the patient and the uncertainty in that impact, so that an attribute of moderate impact can still be ranked critical when little is known about how it behaves. High mannose carries the highest score in the register, which combines a high impact with the largest uncertainty of the glycan attributes. Afucosylation and galactosylation are linked to efficacy through the effector functions of the antibody (ADCC and CDC), and all three glycan attributes are formed in cell culture.

The 2 attributes at the low end of the register are still carried through the campaign. Acidic charge variants (deamidation) and residual DNA are both assigned very low criticality in Table 3.1, and that assignment is an input to this plan and is not revisited here. Both are measured in the studies of the steps that could plausibly move them, and both are tested at release. An attribute of low criticality is not dropped from the assessment, because the assignment itself carries uncertainty and because such an attribute can still show that a step is behaving as the process model says it should (§2).

The form of the criterion follows the kind of risk the attribute carries. 3 attributes carry a two-sided range, because both a low and a high result are a quality concern (the glycan attributes). The process impurities and the size and charge variants carry an upper limit only, since the concern is an elevated level. The 2 viral safety attributes carry a lower limit, expressed as a cumulative log reduction across the process. The acceptance column of Table 3.1 prints an interval for every attribute, so a one-sided criterion appears there with both of its bounds. Only the upper bound of the charge variant criterion is a limit, and for the viral safety attributes only the lower bound is.

Most of the attributes are attributed to the production bioreactor in Table 3.1 (7 of 10), which is why Step 3 carries the largest study of the campaign (§4). The attribution needs qualifying for the process impurities. Host cell protein and residual DNA arise in cell culture, but the level in drug substance is set by the clearance achieved in the purification train, and that clearance is characterized in PCP-005, PCP-007 and PCP-008. Leached Protein A sits differently again. It is attributed in Table 3.1 to the capture step, because it leaches from the resin there, and the parameters that govern how much of it appears are studied in PCP-005.

The 2 viral clearance attributes are cumulative claims, and no single step delivers either of them. In Table 3.1 the enveloped virus claim is attributed to low-pH viral inactivation and the parvovirus claim to anion exchange chromatography. Those attributions name the step that sets the cumulative result, and not the only step that clears virus. The enveloped virus claim is carried by three steps with orthogonal mechanisms, which are the low-pH hold, anion exchange chromatography and small virus retentive filtration (International Council for Harmonisation 2023a). The parvovirus claim is carried by anion exchange chromatography and small virus retentive filtration alone, because a low-pH hold does not inactivate a virus that has no envelope. The plan for each of those steps states the log reduction its own study must support for the attribute that step clears, and both cumulative claims are consolidated in PCMR-001.

The criteria in Table 3.1 are inputs to this campaign. They come from the clinical and non-clinical data package and from prior product experience, and this plan does not re-derive them. Where a study shows that a criterion cannot be met across the ranges studied, the finding is reported in the step's report and the criterion is not adjusted to fit the result.

# 4 Risk-based prioritization

RA-001 sets the scope of the campaign, and this plan does not repeat the assessment. The output of RA-001 is a study type for every parameter, summarized by unit operation in Table 4.1.

Table 4.1: Characterization scope by unit operation. Parameters carried into the campaign, the study-type split assigned by RA-001, and the covering documents.

| Step | Unit operation | Parameters | Multivariate | Univariate | Documents |
|---|---|---|---|---|---|
| 3 | Production Bioreactor | 9 | 5 | 4 | PCP-003 / PCR-003 |
| 4 | Harvest and Clarification | 3 | 0 | 3 | PCP-004 / PCR-004 |
| 5 | Protein A Chromatography | 6 | 4 | 2 | PCP-005 / PCR-005 |
| 6 | Low-pH Viral Inactivation | 4 | 3 | 1 | PCP-006 / PCR-006 |
| 7 | Cation Exchange Chromatography | 5 | 4 | 1 | PCP-007 / PCR-007 |
| 8 | Anion Exchange Chromatography | 5 | 4 | 1 | PCP-008 / PCR-008 |
| 9 | Small-Virus Retentive Filtration | 2 | 2 | 0 | PCP-009 / PCR-009 |
| 10 | Ultrafiltration / Diafiltration | 3 | 0 | 3 | PCP-010 / PCR-010 |

The assessment ranks each parameter twice, once on its potential impact on a quality attribute and once on its potential to interact with other parameters (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b). The two rankings are combined into a score, and the score decides the study type. Where no data or rationale was available for a parameter it was ranked at the highest level, which is the conservative default of the case study. No parameter was excluded from the campaign by the assessment, and the parameters that RA-001 does not link to a quality attribute are studied against the performance of their own step.

Of the 37 parameters carried into the assessment, 22 are assigned to a multivariate design and 15 to univariate study. 6 of the 8 unit operations therefore carry a multivariate design. The other 2 are harvest and clarification (PCP-004) and the UF/DF step (PCP-010). Neither of them governs an attribute in Table 3.1, so their parameters are studied one at a time and are assessed against process performance and against the material each step delivers to the step after it.

The production bioreactor carries the most parameters and the largest multivariate design (5 of 9), which follows from the number of attributes it governs. Small virus retentive filtration carries the fewest parameters, and both go into its multivariate design, because RA-001 expects volumetric load and pressure to act together on virus retention. The capture and polishing steps sit between the two. Each carries a multivariate design in the parameters that govern its clearance and its pool quality, and the remaining parameters are studied univariately.

The assignment is prospective. It reflects what was known before the studies were executed, and it is revisited when the reports are written. A parameter that shows no significant effect is still classified, and it stays in the knowledge space as evidence of robustness. A parameter assigned to univariate study is not moved into a multivariate design during execution without a documented change to its plan.

# 5 Scale-down model strategy

Every characterization study is executed on a scale-down model that has been qualified against the commercial scale process before the study starts. Qualification is performed under SOP-1001 and is reported in the step's report. The model is the warrant for applying a small scale result to commercial manufacture, so each plan states its qualification before it states the study design.

The models are built on the principle that scale-independent parameters (pH, temperature, dissolved oxygen, culture duration) are operated at the commercial set-points, while scale-dependent parameters (agitation, gas flow, pressure, working volume) are set so that the small system matches the performance of the large one. For the production bioreactor this means reproducing the mass transfer and the dissolved carbon dioxide profile of the 15,000 L vessel, and it does not mean geometric similarity to it. In chromatography, bed height, linear velocity, load in grams per litre of resin and all wash and elution volumes in column volumes are kept as they are at commercial scale, and column efficiency is confirmed by plate count and peak asymmetry.

Qualification compares replicate runs at set-point conditions with commercial or pilot scale data, on the attributes the step governs and on step performance. The comparison is quantitative and its acceptance criterion is set in the step's plan, because the attributes and the amount of data available at scale differ from step to step. Where a difference is found, the plan states whether the model is modified or the difference is bounded and carried forward as a limitation on the claims the study can support.

A downstream step is characterized on load material that represents the nominal output of the step before it, so that the effect of the parameters under study is not confounded with variation in the load. Where a study deliberately challenges the load with an extreme input, that challenge is described in the plan and the input is carried as a factor of the design. The load for a given design is prepared under the same conditions across all of its runs, and its identity and quality are confirmed before the study starts, because a shift in the load during execution appears in the analysis as an effect of whatever parameter happened to change with it.

The viral clearance steps need a further model. Spiked runs cannot be performed in the manufacturing facility, so the whole clearance claim rests on a qualified small scale spiking model (International Council for Harmonisation 2023a). Each of the three viral step plans (PCP-006, PCP-008 and PCP-009) states how the spiking model is qualified, how the virus stock is prepared and titred, and how the cytotoxicity and interference controls are run. The limitations of the spiking approach are stated in those plans and are carried into the cumulative claim in PCMR-001.

Each multivariate design includes replicated centre points at the centre of the region studied. They provide the pure error estimate that the lack-of-fit test needs (§6), and they act as a check inside the study that the scale-down system still reproduces its centre-point condition after the extreme runs of the design have been executed.

The scale-down systems are instrumented and are held under calibration and change control. An excursion in calibration status, in a material lot or in the operation of a run is recorded as a deviation, investigated, and assessed for impact in the report of the affected step. The controlled procedures and validated analytical methods the campaign is executed under are listed in Table 5.1.

Table 5.1: Controlled procedures and validated analytical methods under which the characterization campaign is executed. Numbers are placeholders in this synthetic package.

| Reference | Title | Type |
|---|---|---|
| SOP-1001 | Qualification of Scale-Down Models | SOP |
| SOP-1002 | Design, Execution and Statistical Analysis of DoE Studies | SOP |
| SOP-2003 | Operation of the Production Bioreactor (Fed-Batch) | SOP |
| SOP-2004 | Cell-Culture Media and Feed Preparation | SOP |
| SOP-2005 | Harvest by Continuous Disk-Stack Centrifugation | SOP |
| SOP-2006 | Clarification by Depth and Sterile Filtration | SOP |
| SOP-2007 | Operation of the Protein A Capture Chromatography Step | SOP |
| SOP-2008 | Chromatography Resin Life-Cycle, Packing and Sanitization | SOP |
| SOP-2009 | Operation of the Low-pH Viral Inactivation Step | SOP |
| SOP-2010 | Operation of the Cation-Exchange Polishing Chromatography Step | SOP |
| SOP-2011 | Operation of the Anion-Exchange Polishing Chromatography Step | SOP |
| SOP-2012 | Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step | SOP |
| SOP-2013 | Operation of the Ultrafiltration / Diafiltration (Formulation) Step | SOP |
| SOP-3010 | Released N-Glycan Mapping by 2-AB HILIC-UPLC | SOP |
| SOP-3011 | Size-Variant Analysis by SEC-HPLC | SOP |
| SOP-3012 | Host-Cell-Protein Quantitation by ELISA | SOP |

| Reference | Title | Type |
|---|---|---|
| SOP-3013 | Charge-Variant Analysis by icIEF | SOP |
| SOP-3014 | Residual Host-Cell DNA by qPCR | SOP |
| SOP-4001 | Process-Parameter Classification and Control-Strategy Definition | SOP |
| AMV-3010 | N-Glycan Map (2-AB HILIC-UPLC) | Method validation |
| AMV-3011 | Size-Variants (SEC-HPLC) | Method validation |
| AMV-3012 | Host-Cell Protein ELISA | Method validation |
| AMV-3013 | Charge Variants (icIEF) | Method validation |
| AMV-3014 | Residual DNA (qPCR) | Method validation |
| AMV-3015 | Turbidity by Nephelometry (NTU) | Method validation |
| AMV-3016 | Leached Protein A by ELISA (ppm) | Method validation |
| AMV-3017 | XMuLV Infectivity Titre (TCID50) | Method validation |
| AMV-3018 | MVM Infectivity Titre (TCID50/qPCR) | Method validation |
| AMV-3019 | Protein Concentration by A280 (UV) | Method validation |

A qualified scale-down model supports conclusions about the effect of a parameter and about the width of a range. It does not demonstrate commercial scale performance on its own. Confirmation at scale is a Stage 2 activity, and each report states that boundary where it states a capability.

# 6 Common statistical approach

One analysis method is used at every step that has a multivariate design, so that results can be compared across the train. The design is executed in two stages. A two-level screening design identifies which parameters affect each response, and a response-surface design in the parameters carried forward gives the predictive model. All analyses are performed under SOP-1002.

Factors are coded before analysis, so that effects can be compared between parameters measured in different units. The coding is set by the range studied and not by the set-point. The centre of the range is the coded level 0 and its edges are the coded levels −1 and +1. The set-point lies inside the range for every parameter of the campaign, and it lies at the centre of the range for 16 of the 22 parameters carried into a multivariate design. An effect is reported as the change in the response across the whole range studied, which is twice the coded coefficient.

The screening design is a two-level factorial in the parameters assigned to multivariate study, augmented with replicated centre points. A full factorial is used where four or fewer parameters are carried. Where five are carried, a Resolution V half fraction is used, which keeps the main effects and the two-factor interactions unaliased with each other. The model fitted to the screening data carries main effects and two-factor interactions, it is fitted by ordinary least squares (OLS) on the coded factors, and terms are judged at $\alpha = 0.05$.

The response-surface design is a face-centred central composite design in the parameters carried forward from screening. Its axial points sit on the faces of the cube, so every run stays inside the range the plan justified. The fitted model is a full quadratic. The screening model identifies the active parameters, and the response-surface model is the predictive model and the basis of the design space. The screening design is close to saturated, and its fit is not used to predict.

The parameters assigned to univariate study are assessed one at a time across their range, with the other parameters of the step held at their set-points. The response is compared with its criterion and the range over which the criterion is met is reported as the proven acceptable range for that parameter. A univariate study cannot detect an interaction, so a parameter is assigned to one only where prior knowledge and the mechanism of the step give no reason to expect that it interacts with the parameters in the multivariate design (§4).

Responses that describe process performance, such as step yield, are modelled in the same way as the quality attributes. They are judged against the performance expected of the step and not against a drug substance criterion, and a parameter that moves a performance response without moving a quality attribute is classified as a key process parameter (§7).

Model adequacy is judged on four criteria, and all four are reported. The overall regression must be significant. The adjusted coefficient of determination and the predicted coefficient of determination, which comes from the PRESS statistic, must agree closely enough to show that the model is not merely fitting its own data. The ANOVA lack-of-fit test against the pure error from the centre point replicates must not be significant. The residual diagnostics (residuals against predicted values, a normal probability plot, and observed against predicted values) must show no structure. A model that fails the lack-of-fit test is reported as descriptive and is not used to predict, because a model that cannot reproduce its own centre points within the pure error of the assay gives no assurance at the edges of the region, which is exactly where the design space is set.

Proven acceptable ranges are derived from the fitted response-surface model, and two analyses are reported for each parameter that governs an attribute. The first scans the parameter across the range studied with the other parameters of the design held fixed, and each plan states the settings it uses. The second propagates the variation of the other parameters within their normal operating ranges by Monte-Carlo, which is the analysis the control strategy relies on. Both are presented with the acceptance limit drawn, the acceptable region shaded, and the set-point and the normal operating range marked.

Capability is estimated by simulation. 2,000 commercial scale batches are simulated with every parameter drawn about its set-point, from a normal distribution whose standard deviation is one sixth of the width of the normal operating range and which is truncated at the edges of the range studied. The resulting distribution of each drug substance attribute is compared with its acceptance criterion. Capability is reported as a one-sided capability index (Cpk) for the impurities and for viral clearance, where only one limit is at risk, and against both limits where the criterion is two-sided.

The capability estimate describes the process as represented by the qualified scale-down models, with every parameter inside its normal operating range. It does not carry commercial scale variation that the models do not reproduce, and it is confirmed against the process performance qualification (PPQ) batches in Stage 2.

# 7 Acceptance criteria framework

Each characterization plan states its acceptance criteria before its studies are executed. This section gives the framework those criteria instantiate. Where a threshold depends on the step, this plan gives the form of the criterion and the plan for the step sets the value.

## 7.1 A qualified scale-down model

A model is qualified when its performance at set-point conditions is comparable to the commercial or pilot scale process, both for the attributes the step governs and for step performance. Comparability is assessed statistically on replicate runs, and the numeric limit is set in the step's plan. A model that is not qualified cannot support a range claim, so the study is not executed until the qualification is complete and approved.

## 7.2 An adequate model

A model is adequate when it meets the four criteria of §6. The overall regression is significant, the adjusted and predicted coefficients of determination agree, the lack of fit is not significant against pure error, and the residuals show no structure. The thresholds are stated per study, because the responses differ in precision from one assay to another. A response for which no parameter is active is a valid outcome and is reported as one. It is presented with the effect estimates and their significance, and the parameters are then classified on that evidence.

## 7.3 An acceptable operating region

An operating region is acceptable when every attribute the step governs stays inside its acceptance criterion across the region, including at the worst-case corner of the region. For a formed attribute or an impurity the criterion is the drug substance criterion in Table 3.1. For a viral clearance attribute the criterion is the step's own required log reduction, back-calculated from the cumulative requirement less the clearance credited to the other steps. No step is asked to carry the whole viral claim and no clearance is credited twice. Movement inside the design space is not a change to the process from the point of view of the regulatory filing (International

Council for Harmonisation 2009), although every planned move is assessed under the pharmaceutical quality system before it is made, and that assessment records the risk of the particular move and the knowledge that supports it (International Council for Harmonisation 2008). Movement outside the design space requires change control, and a submission where the registered ranges are affected.

## 7.4 Parameter classification

Every parameter is classified in its step's report under SOP-4001, and the classification is the input to the control strategy. A parameter with a demonstrated effect on a critical quality attribute is a critical process parameter. It is called well controlled (WC-CPP) where the equipment and the control system hold it comfortably inside the design space, and critical (CPP) without that qualification where the risk of leaving the design space is material (CMC Biotech Working Group 2009). A parameter that affects process performance but no quality attribute is a key process parameter (KPP). A parameter with no demonstrated effect on either is a general process parameter (GPP), and the evidence for the absence of an effect is stated with the classification and not merely implied by it.

This plan sets no campaign-wide numeric threshold for capability. Each report states the capability achieved for each attribute together with the margin to the limit, and any threshold agreed for a particular step is stated in that step's plan. A capability figure is not by itself evidence that the process is in a state of control. That is established over the commercial batches of Stage 2 and the continued process verification that follows them.

# 8 Register of characterization plans

The characterization plans this master plan governs are listed in Table 8.1. Each is approved before the studies it covers are executed, and each is paired with the report that presents those studies. A plan is revised under change control where its design changes before execution, and the revision is approved before the affected runs are executed. The study type each plan covers follows the assignment in Table 4.1.

Table 8.1: Register of the characterization plans governed by this master plan, with their paired reports.

| Plan | Subject | Paired report |
|---|---|---|
| PCP-003 | Production Bioreactor (Step 3) | PCR-003 |
| PCP-004 | Harvest and Clarification (Step 4) | PCR-004 |
| PCP-005 | Protein A Chromatography (Step 5) | PCR-005 |
| PCP-006 | Low-pH Viral Inactivation (Step 6) | PCR-006 |
| PCP-007 | Cation Exchange Chromatography (Step 7) | PCR-007 |
| PCP-008 | Anion Exchange Chromatography (Step 8) | PCR-008 |
| PCP-009 | Small-Virus Retentive Filtration (Step 9) | PCR-009 |
| PCP-010 | Ultrafiltration / Diafiltration (Step 10) | PCR-010 |

# 9 Deliverables and schedule

The campaign delivers this plan, 8 characterization plans, the executed study data, 8 characterization reports and the master report PCMR-001. Each report is issued after the studies in its plan are complete and analysed. PCMR-001 is issued after the last report and consolidates them.

The sequence is the same at each step that carries a multivariate design. RA-001 assigns the study type, the plan is approved, the scale-down model is qualified, and the screening study is executed. The response-surface study follows in the parameters that screening identified, and the univariate studies run in parallel with both. At harvest and clarification and at UF/DF the univariate studies stand alone, and the same approval and qualification precede them. Analysis, parameter classification and the report close the step in either case. The campaign itself closes when PCMR-001 is approved, which is the point at which the 5 process performance qualification batches of Stage 2 can be planned against a defined control strategy.

No calendar dates are fixed in this plan. The schedule of the campaign is managed under PTP-001 together with the transfer milestones.

# 10 Approvals

This plan takes effect when it is approved by the functions listed in Table 1. It is revised as a whole document under site change control. A change to the quality attribute framework, to the risk basis or to the acceptance framework requires each affected characterization plan to be reviewed for impact before its studies proceed.

# References


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